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High speed laser scanning microscopy by iterative learning control of a galvanometer scanner

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ABSTRACT

Iterative learning control (ILC) for a galvanometer scanner is proposed to achieve high speed, linear, and accurate bidirectional scanning for scanning laser microscopy. A galvanometer scanner, as a low stiffness actuator, is first stabilized with a feedback control compensating for disturbances and nonlinearities at low frequencies, and ILC is applied for the control of the fast scanning motion. For stable inversion of the non-minimum phase zeros, a time delay approximation and a zero phase approximation are used for design of ILC, and their attainable bandwidths are analyzed. Experimental results verify the benefits of ILC of its wide control bandwidth, enabling a faster, more linear, and more accurate scanning without a phase lag and a gain mismatch. At the scan rate of 4112 lines per second, the root mean square (RMS) error of the ILC can be reduced by a factor of 73 in comparison with the feedback controlled galvanometer scanner of the commercial system.

1. Introduction

Galvanometer based scanning mirrors are the most widely used scanning systems for high-precision in vivo biological imaging systems, such as scanning confocal and two-photon excitation microscopes (Aylward, 2003; Pawley, 2006; Saggau, 2006). In principle, these microscopes record an image by scanning a laser point by point to achieve a high spatial resolution. A high temporal resolution of microscopic images is desirable for capturing rapid biological phenomena such as a cardiac cycle (Baader, Buchler, Bircher-Lehmann, & Kleber, 2002) or tracking the single molecule movements in living organisms (Meijering, Smal, & Danuser, 2006). Fast imaging in scanning laser microscopes, however, is challenging in practice due to dynamics of the scanner, limiting the bandwidth of the controller (Duma, Lee, Meemon, & Rolland, 2011; Ji, Shroff, Zhong, & Betzig, 2008). To minimize the imaging time and increase the temporal resolution, fast and accurate scanning is a crucial requirement for galvanometer scanners.

One approach to increase the imaging speed is bidirectional scanning (Duma, 2010), doubling the scan rate by using both trace and retrace. Fig. 1 shows two ways of scanning: conventional unidirectional scanning based on sawtooth scanning (left) and bidirectional scanning based on triangular scanning (right), respectively. For conventional scanning laser microscopes, sawtooth scanning is commonly used for unidirectional scanning due to the long linear slope for imaging (Duma et al., 2011; Trepte & Liljeborg, 1994). However, sawtooth scanning is inefficient due to the small duty cycle by long and fast retracing, which may cause overheating of the actuator (Podoleanu & Nicolov, 2009; Montagu et al., 2011) and require an additional shutter for preventing unnecessary photobleaching of the specimen (Pawley, 2006). For bidirectional scanning of galvanometer scanners, Duma et al. (2009), Duma and Awrejcewicz (2009), and Duma (2010) intensively studied various types of bidirectional scanning trajectories for a large linear scan area and compared them to optical coherence tomography (Duma et al., 2011). They point out that triangular scanning is desirable comparing to unidirectional scanning, but it is difficult to obtain an accurate scan at a high scanning speed due to high frequencies induced by the sharp turnaround. Chen et al. also point out that fast bidirectional scanning suffers from mismatches between the trace and retrace due to the phase lags and nonlinearities, deteriorating the quality of the images (Chen et al., 2010). A manufacturer of scanning laser microscopes provides a bidirectional scan option with a phase correction control bar for users to compensate for this mismatch manually (Leica Microsystems, 2005). However, the vibrational dynamics of the scanner, which causes positioning errors near the turnaround, is completely ignored so far.

As a solution to compensate for these drawbacks of the triangular scanning motion, iterative learning control (ILC) can be applied. ILC is a feed-forward control for tracking a repetitive reference by updating the control signal based on tracking error measurements obtained at previous trials. For fast scanning ILC has been applied to atomic force microscopy (AFM) (Leang &

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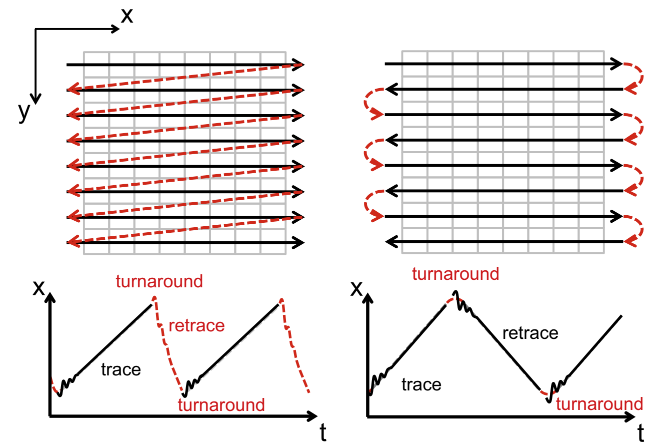


Fig. 1. Unidirectional scanning (left top) based on sawtooth scanning for the fast x-axis over the time t (left bottom) and bidirectional scanning (right top) based on triangular scanning (right bottom) for the fast x-axis. The focused laser beam is moved linearly over the specimen (gray squares) for recording an image along the active scanning region (black solid line). At a high scan rate, both sawtooth and triangular scanning of the fast x-axis may suffer from distortion of trace and retrace scans (black solid line) compared to the desired reference line (gray dotted line).

Devasia, 2006; Schitter, 2009; Tien, Zou, & Devasia, 2006) and an optical scanning system (Yen, Yeh, Peng, & Lee, 2009) to compensate for the dynamics and nonlinearity of the high stiffness piezoelectric actuators. ILC is applied for a galvanometer scanner for a microvia drilling machine to achieve a high tracking and settling performance for integrated circuit correction (Potsaid, Wen, Unrath, Watt, & Alpay, 2007; Wen & Potsaid, 2004). For confocal laser scanning microscopy, an iterative optimization concept, which corresponds to an I type ILC, have been proposed for a 50 Hz unidirectional scan with 85% active linear scan region, reporting a RMS error 0.06% in the linear region (Trepte, 1996). However, compensation of the vibrational modes due to the internal modes of the galvanometer scanner for a fast and accurate scanning over 1 kHz in optical microscopy has not been studied yet.

The goal of this paper is to develop ILC for bidirectional scanning beyond 1 kHz, i.e. 2000 lines per seconds, to be applied to fast and distortion-free laser scanning microscope imaging (Yoo, Ito, Verhaegen, & Schitter, 2012). Section 2 describes the stabilized galvanometer scanner, consisting of the galvanometer scanner and a feedback controller, and its linear model is derived. Based on the model of the stabilized galvanometer scanner, ILC design is investigated in Section 3 and two approaches for deriving a stable inversion of the non-minimum phase scanner dynamics are presented. Experimental results verify the benefits of the proposed ILC approach in Section 4, and conclusions are drawn in Section 5.

2. Stabilized galvanometer scanner

2.1. Galvanometer scanner

For the experiments, a high performance galvanometer scanner (6210 H, Cambridge Technology Inc., Lexington, MA, USA) is used, which is widely installed in commercial laser confocal microscopes (Aylward, 2003; Pawley, 2006). The galvanometer scanner is a moving-magnet type as shown in Fig. 2, i.e. magnets along the shaft are rotated by the Lorentz force created by the current i_c through the fixed coils. A scanning mirror is attached at one end of the shaft for directing a laser beam. At the other end of the shaft an encoder is attached for the precise angle measurement of the mirror by the passed light through the blocking butterfly (Aylward, 2003). The mechanical actuation range θ_m of the galvanometer scanner is $\pm 10°$, corresponding to $\pm 20°$ in optical angle θ_o. The servo driver (MicroMax 671, Cambridge Technology Inc.) consists of a current driver for the coil of the galvanometer scanner, the decoder for the encoder signal, and a closed loop control. The system input and the system output are given by the input to the current driver and the encoder signal through the decoder, respectively. The closed loop control of the servo driver is only used as a benchmark for comparison with the ILC in Section 4, but is disabled for all other experiments.

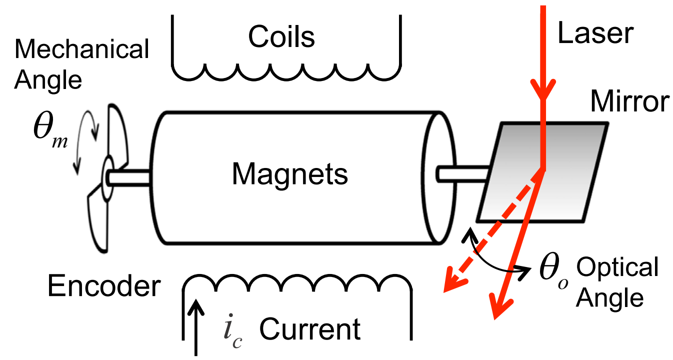


Fig. 2. Structure of a moving-magnet galvanometer scanner. The rotor of a galvanometer scanner consists of magnets, a mirror and an encoder connected by a steel shaft, and coils surrounding the magnets. The Lorentz force induced by the coil current i_c with magnets rotates the mirror at one end and the encoder at the other end measures the mirror angle, mechanical angle θ_m. The deflected mirror angle change the angle of the laser, i.e. optical angle $\theta_o = 2\theta_m$.

The galvanometer scanner is marginally stable since the rotor of the galvanometer can be regarded as a floating mass with a weak stiffness and friction at low frequency. Fig. 3 shows the frequency response of the galvanometer $G(s)$ from the control input to the encoder output measured by a Dynamic Signal Analyzer (HP3562, Agilent Technologies, Santa Clara, CA, USA). The frequency response shows the dynamics between 300 Hz and 30 kHz dominated by the inertia of the rotor, providing a -40 dB/dec line in Fig. 3. At low frequencies, however, the magnitude does not show a slope of -40 dB/dec due to the friction, stiction, and weak suspension springs. To counteract the marginal stability and drift of low frequencies, a feedback controller is designed and

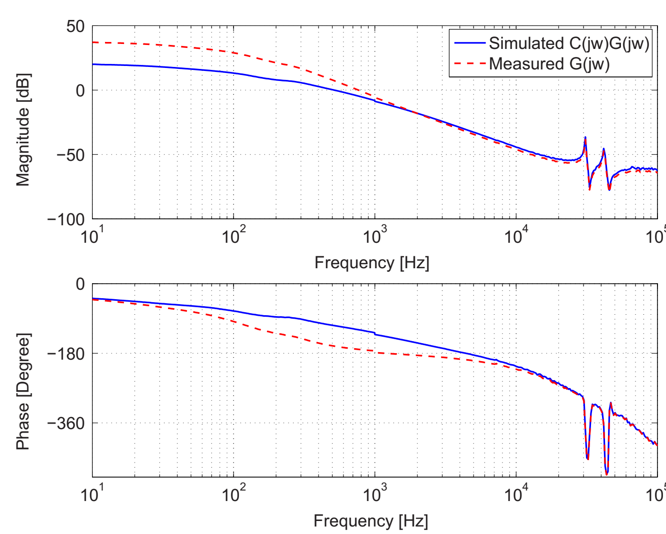


Fig. 3. Measured Bode plot of the galvanometer scanner from the system input to the system output (red dash line) and the simulated frequency response of open-loop transfer function with a tamed PD controller $C(jw)P(jw)$ (blue solid line). (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

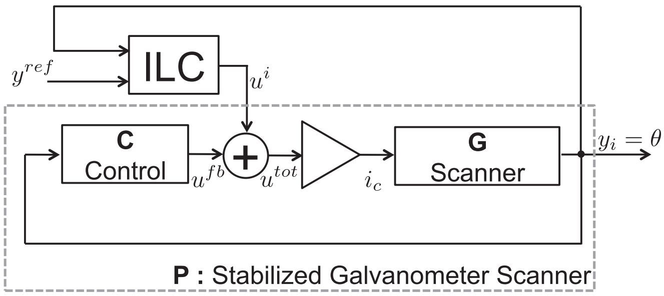


Fig. 4. Overall architecture of the system with ILC. The galvanometer scanner G is stabilized by the feedback controller C and forms a stabilized galvanometer-based scanning system P, which is the process sensitivity $P(s) = G(s)/(1 + G(s)C(s))$. The control signal from ILC u^i for high frequency scanning is summed with the control signal from the feedback controller u^{fb}, resulting in the total input u^{tot} for the galvanometer. The total input, $u^{tot} = u^{fb} + u^i$, is applied to the current driver.

implemented before ILC is applied.

2.2. Feedback control design for stabilization

Fig. 4 illustrates the overall structure of the control system, where G, C, and ILC blocks represent the galvanometer scanner, a stabilizing feedback controller, and the iterative learning controller, respectively. The ILC is applied in parallel to the servo driver with the stabilizing feedback control by an analog summing amplifier with 1.5 MHz bandwidth (-3 dB). This structure allows a broad bandwidth of the ILC, regardless of the stabilizing feedback controller. Before the ILC design a PD feedback controller is designed and implemented for stabilizing the marginally stable galvanometer scanner.

For the stabilization of the galvanometer scanner, a lead compensator, i.e. a tamed PD controller, is selected as the feedback controller $C(s)$ for providing a phase margin around the unit gain crossover frequency (Mnerie, Preitl, & Duma, 2013; Schmidt, Schitter, Rankers, & van Eijk, 2014). The controller is tuned, such that the crossover frequency of the open-loop transfer function $G(s)C(s)$ is about 500 Hz with a sufficient phase margin as shown in Fig. 3. The simulation shows that the gain of $G(s)C(s)$ crosses the 0 dB line at 500 Hz with a phase margin of 72° and the gain margin is 36 dB. The controller is implemented as an analog circuit with operational amplifiers and passive components.

Note that the feedback controller is intentionally tuned to have a large gain and phase margin, resulting in a relatively low control bandwidth. With this setting, uncertainties at low frequencies such as a deflection dependent friction and drift are compensated by the feedback control. In addition, the sensor noise, which gets more dominant with high frequencies, stays low in the stabilized galvanometer scanner (Skogestad & Postlethwaite, 2005). This is beneficial for the ILC design since it makes the dynamics of the stabilized galvanometer scanner more deterministic. Based on this, ILC is applied to the stabilized galvanometer scanner to control the fast scanning motion at high frequencies.

2.3. Linear system modeling

For the design of inversion based ILC, the stabilized galvanometer scanner model $P(s)$ from the ILC input to the encoder output is necessary. Fig. 5 illustrates a measured Bode plot of the encoder output of the stabilized galvanometer scanner from the control input by the Dynamic Signal Analyzer (HP3562, Agilent Technologies) and the Bode plot of its manually fitted linear model. The manually fitted linear model is obtained based on multiple second order systems by adjusting each natural frequency and damping parameter in Matlab (The MathWorks, Inc., Natick, Massachusetts, USA). The transfer function from the

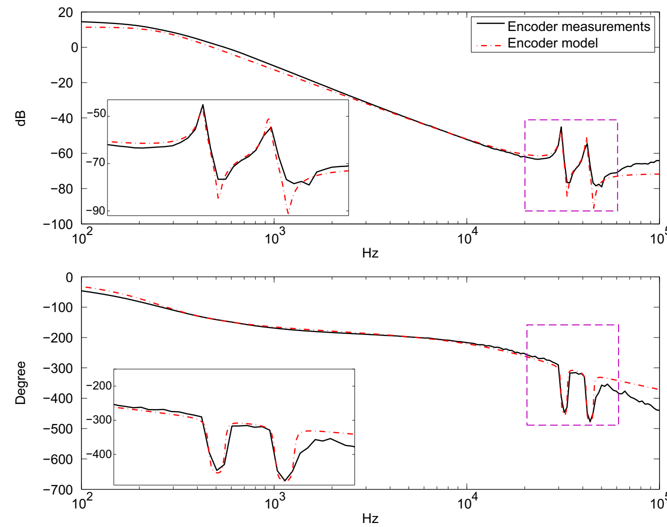


Fig. 5. Measured Bode plots of the encoder output (solid black line) of the stabilized galvanometer scanner from the input for ILC (u^i) and the Bode plot of the linear model with non-minimum phase zeros (dash-dotted red line). The zoomed Bode plots (violet dashed box in original Bode plot) show the resonance and anti-resonances of the galvanometer scanner dynamics.

system input, given by the current through the coils of the galvanometer, to the encoder output is given as follows (Yoo et al., 2012):

$$\hat{P}(s) = \frac{K\prod_{iz=1}^{n_z}(s^2 + 2\zeta_{iz}\omega_{iz}s + \omega_{iz}^2)}{\prod_{ip=1}^{n_p}(s^2 + 2\zeta_{ip}\omega_{ip}s + \omega_{ip}^2)}e^{-t_d s}, \tag{1}$$

where K denotes a gain, t_d is a time delay. n_p and n_z denote the number of pole and zero pairs, respectively.

Table 1 provides all parameters of the transfer function in Eq. (1), and its Bode plot is shown by the red dashed-dot line in Fig. 5. The table shows that a pair of non-minimum phase zeros occurs at 25 kHz. Fig. 6 shows a measured transient response and simulation results for the reference input of a square function with an amplitude of 0.6 V. The measured transient responses are averaged by 322 cycles for reducing the measurement noise. Undershoots in both rising and falling edges indicate the presence of non-minimum phase zeros in the stabilized galvanometer scanner (Hoagg & Bernstein, 2007).

These non-minimum phase zeros provide a fundamental limit of the bandwidth of the controller in practice. For feedback control, the bandwidth of the control is limited by the frequency of non-minimum phase zeros (Hoagg & Bernstein, 2007; Skogestad & Postlethwaite, 2005). Non-minimum phase zeros are also problematic in the feedforward control design such as ILC, but several stable design techniques are available to overcome the limit (Cai, Freeman, Lewin, & Rogers, 2008; Devasia, Chen, & Paden, 1996; Potsaid et al., 2007; Rigney, Pao, & Lawrence, 2009; Tomizuka, 1987). By such design techniques ILC can achieve a wide control bandwidth for high tracking performance in fast beam scanning. In the following section ILCs and stable inversion methods are

Table 1
Parameters of the stabilized galvanometer scanner model.

Poles		Zeros		t_d	K
$\omega_{ip}/2\pi$	ζ_{ip}	$\omega_{iz}/2\pi$	ζ_{iz}	(μs)	
250	0.65	25,000	-0.85		
30,800	0.0087	33,200	0.0069	1	2.64×10^{-4}
41,700	0.008	45,500	0.0084		

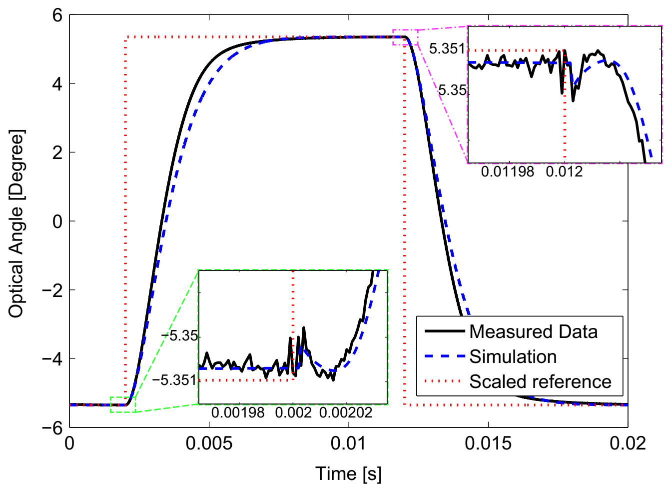


Fig. 6. Transient responses of a measured encoder output (black solid line) and simulation (blue dashed line) for the reference input of a square function (red dotted line). The reference input is scaled for better readability of the transition time. For the rising edge (green dashed square) and falling edge (violet dashed-dot square) are zoomed to observe an undershoot by non-minimum phase zeros.

discussed for the fast and accurate scanning of the galvanometer scanner.

3. Iterative learning control design

In this section a design of ILC is discussed for the fast and accurate scanning motion control based on the linear model of the galvanometer scanner. ILC is beneficial for scanning motion control since non-unity magnitude and phase lags at each harmonics can be corrected by its learning nature and periodic disturbances and nonlinearities can be compensated for over a wide bandwidth. In the first half of this section, two inversion based ILCs are discussed for the linear model with non-minimum phase zeros.

3.1. Inversion based iterative learning control with non-minimum phase zeros

Fig. 7 shows a schematic diagram of the iterative learning control with a learning filter L and a Q-filter (Bristow, Tharayil, & Alleyne, 2006). The structure leads to the following equation as :

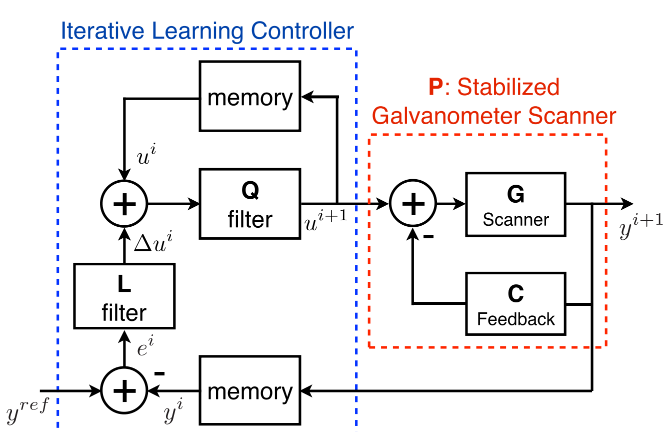


Fig. 7. A schematic diagram of the iterative learning controller. A learning filter L calculates a required update Δu^i based on the error e^i between the previous system output y^i and the reference trajectory y^{ref}. The update Δu^i is added to the current ILC input u^i, which is saved in the memory, and used as a new ILC u^{i+1} in the next iteration after a robustness filter Q is applied.

$$u_k^{i+1} = Q(q)\left(u_k^i + L\left(q\right)e_k^i\right), \tag{2}$$

$$e_k^i = y_k^{ref} - y_k^i, \tag{3}$$

$$y_k^i = P(q)u_k^i, \tag{4}$$

where u_k^i, y_k^i and e_k^i denote the ILC input, output, and tracking error at the discrete time k in ith trial. $L(q)$, $Q(q)$, and $P(q)$ denote the LTI dynamics of the learning filter, Q-filter, and the plant with the forward time-shift operator q, i.e. $x(k+1) = qx(k)$. y_k^{ref} is the desirable scanning trajectory at time k.

Then the error evolution equation in the frequency domain is as follows:

$$E^{i+1}(z) = Q(z)\left(1 - P(z)L(z)\right)E^i(z) + \left(1 - Q(z)\right)Y^{ref}(z), \tag{5}$$

where $Y^{ref}(z)$ and $E^i(z)$ are the z-transform of the reference trajectory and of the error at ith trial, respectively, with $z = e^{jwT_s}$ with sampling time T_s. This provides two well-known characteristics of ILC. The first is that the straightforward solution for the learning filter design is the inversion of the process dynamics $P(z)$ to make the error zero (Bristow et al., 2006). A learning filter $L(z)$ is designed with a learning gain $\rho \in \mathbb{R}^+$, multiplied with an inverse dynamics, i.e. $L(z) = \rho\hat{P}^{-1}(z)$, where $\hat{P}(z)$ denotes the system dynamics model. As the learning gain gets smaller, the robustness to the modeling errors is improved (Harte, Hätönen, & Owens, 2005), albeit sacrificing on the learning speed. The second is that the ILC algorithm converges if the maximum spectral radius of the error evolution operator in trial axis is smaller than 1, which leads to a following renowned ILC convergence criterion (Norrlöf & Gunnarsson, 2002),

$$\left|Q(z)\left(1 - P(z)L(z)\right)\right| < 1, \quad \forall\, w. \tag{6}$$

From Eq. (6), $Q(z)$ denotes a robustness filter, also called Q-filter, which is usually designed to suppress the modeling and design error of L compared to the actual dynamics P. For w that satisfies $Q(w) = 1$, Eq. (6) can be rewritten as (Fleming & Leang, 2014)

$$\rho\left|\frac{P(w)}{\hat{P}(w)}\right| < 2\cos\left(\arg(\hat{P}(w)) - \arg(P(w))\right), \tag{7}$$

where $\arg(\cdot)$ denotes the phase of the complex value. Since magnitude is nonnegative, Eq. (7) leads to a following condition as (Tien et al., 2006)

$$\left|\arg(\hat{P}(w)) - \arg(P(w))\right| < \arccos\left(\frac{\rho}{2}\left|\frac{P(w)}{\hat{P}(w)}\right|\right) \leq \frac{\pi}{2}. \tag{8}$$

Eq. (8) means that the uncertainty and modeling error of the system dynamic model should be less than $\pm 90°$ in phase for the convergence of ILC. For this reason modeling errors have to be kept small in phase as well as in magnitude.

In this straightforward and inversion based design of the learning filter $L(z)$, one of the major challenges lies on the stable inversion of the plant model $\hat{P}^{-1}(z)$ when the plant has non-minimum phase zeros. Fig. 8 shows pole-zero plots of the discretized stabilized galvanometer scanner model (black x and o marks at the locations of poles and zeros, respectively), and the non-minimum phase zeroes at 25 kHz (black o marks in black dotted line box) are observed outside of the unit circle. Direct inversion of such non-minimum phase zeros would lead to an unstable learning filter. Among the stable inversion methods (Rigney et al., 2009), zero phase approximation (Tomizuka, 1987;

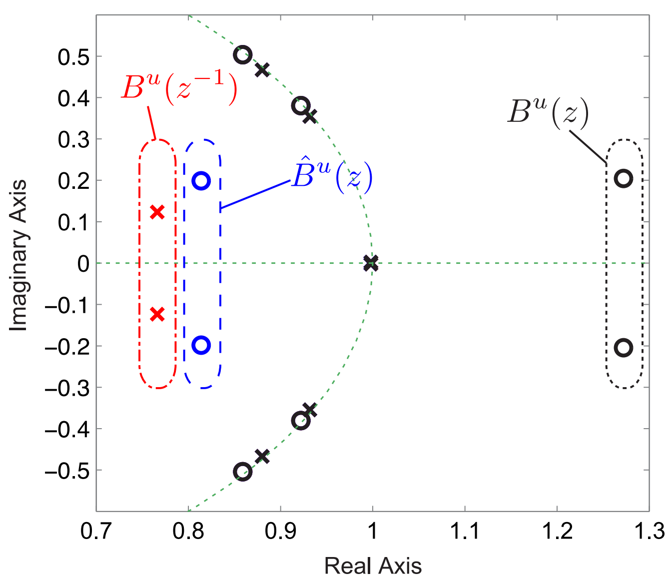


Fig. 8. Pole-zero plot of the stabilized galvanometer scanner model $\hat{P}(z)$ with non-minimum phase zeros ($B^u(z)$, black dotted line box), zero phase approximation $\hat{P}_{zp}(z)$ with stable poles ($B^u(z^{-1})$, red dashed-dot line box) and an invertible model with minimum phase zeros ($\hat{B}^u(z)$, blue dashed line box) for the delay approximation $\hat{P}_{td}(z)$.

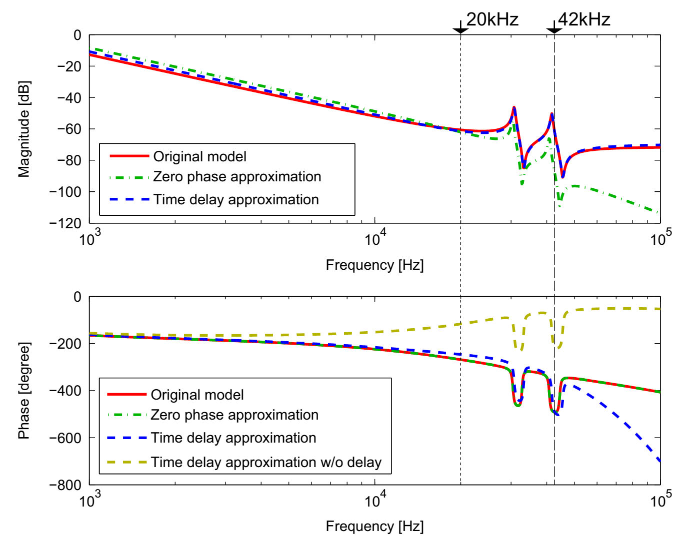


Fig. 9. Bode plots of a model with non-minimum phase zeros (red solid line), zero phase approximation of ZPETC (green dashed-dot line), delay approximation method without time delay (dark yellow dashed lines in phase diagram) and with time delay (blue dashed line). The arrows of 20 and 42 kHz represent the cutoff frequency of the ILC designs with the zero phase and delay approximations, respectively.

Wen & Potsaid, 2004) and time delay approximation (Cai et al., 2008, Potsaid et al., 2007) are chosen due to their lower complexity at implementation (Butterworth, Pao, & Abramovitch, 2012), which requires a linear filter and a memory for the time shift. These two inversion methods are discussed for a stable L filter design for ILC in the following subsections.

3.1.1. Zero phase (ZP) approximation

To cope with non-minimum phase zeros, the idea of zero phase error tracking control (ZPETC) can be used for a stable inversion (Tomizuka, 1987). Assume that the numerator of the plant model can be factorized by the terms of minimum phase zeros $B^s(z)$ and non-minimum phase zeros $B^u(z)$, i.e.

$$\hat{P}(z) = \frac{B^u(z)B^s(z)z^{-k_d}}{A(z)}, \tag{9}$$

with the denominator $A(z)$ and the time delay in discrete time $k_d = \text{round}(t_d/T_s)$, where round$(\cdot)$ denotes the round-off of the time delay into the integer delay in discrete time. Then a zero phase approximation of the stable model inversion is given as follows (Tomizuka, 1987):

$$\hat{P}_{zp}^{-1}(z) = \frac{A(z)B^u(z^{-1})z^{k_d}}{B^s(z)\left(B^u(1)\right)^2}. \tag{10}$$

By the inversion the time delay is changed into the time advance. At the implementation this time advance simply can be applied by shifting the input signal of the next trial or the output signal of the previous trial. In this paper, the increment of the input for the update is shifted for the realization of the time advance.

In Fig. 8 this approximation can be observed by minimum-phase poles (red x marks in red dashed-dot line box) in lieu of original non-minimum phase zeros. This leads to the zero phase error in the model inversion, however, the magnitude error of the model inversion increases at the rate of 40 dB per decade from the frequency of the pair of non-minimum phase zeros onwards. The Bode plot of the zero phase approximation (green dashed-dot line) in Fig. 9 shows the identical phase with the original model with the non-minimum phase zeros (red solid line) while the magnitude does not fit anymore at frequencies higher than the replaced non-minimum phase zeros. This magnitude mismatch can violate the convergence criterion (6) of the ILC and mainly limits the bandwidth of the ILCs, degrading the tracking performance.

3.1.2. Time delay (TD) approximation

Another approach to treat the non-minimum phase zeros for a stable model inversion is a time delay approximation (Åström & Hägglund, 2006; Potsaid et al., 2007). The system with non-minimum phase zeros can be approximated by a system with minimum phase zeros and a time delay as follows (Butterworth, Pao, & Abramovitch, 2012; Cai et al., 2008; Potsaid et al., 2007):

$$\hat{P}_{td}^{-1}(z) = \frac{A(z)z^{k_d+k_a}}{B^s(z)\hat{B}^u(z)}, \tag{11}$$

where $\hat{B}^u(z)$ is an approximated minimum phase zeros (blue o marks in blue dashed line box) instead of non-minimum phase zeros as shown in Fig. 8, and k_a denotes the discretized delay of an additional input delay t_a for the phase correction, i.e. $k_a = \text{round}((t_d + t_a)/T_s) - k_d$. This inversion by stable zeros can provide a small magnitude error as shown in Fig. 9 (blue dashed line) (Åström & Hägglund, 2006; Wen & Potsaid, 2004). In contrast to the zero phase approximation, the phase error can be large without delay (dark yellow dashed line in Fig. 9) which may cause divergence of the ILC by Eq. (8). By adding a time delay to the approximated model the phase lead due to the stable zeros can be compensated up to a certain frequency, providing a good approximation of the model in a wide frequency range (blue dashed line in Fig. 9). At high frequencies beyond the ILC bandwidth the phase lag starts to diverge from the dynamics of the galvanometer.

In practice $\hat{B}^u(z)$ and t_d are determined by the trade off between the magnitude and the phase error of the model inversion for satisfying the convergence criteria of Eqs. (6) and (8). For high performance of ILC, $\hat{B}^u(z)$ can be initially chosen by $B^u(z^{-1})$ for zero magnitude error (Butterworth et al., 2012; Wen & Potsaid, 2004) and further fine-tuned referring to the frequency responses for a better fit. Then the time delay is chosen to compensate for the phase lead due to the transmission zeros (Åström & Hägglund, 2006). Compared to the optimization (Cai et al., 2008) and online adaptation (Butterworth et al., 2012) of the time sequences and to system identification with a delay (Potsaid et al., 2007), the delay in this case is simply chosen by frequency responses in Fig. 9 to satisfy Eq. (8) up to the desired frequency, which is an upper bound of the control bandwidth of the ILC. As a result, the time delay approximation in Figs. 8 and 9 is obtained by replacing damping parameter -0.85 with 0.6 for the non-minimum phase zeros at 25 kHz and the total input delay, $t_d + t_a$, is chosen as 18.5 μs, which corresponds to a maximum 30° phase error before the zero phase error crossing at 36 kHz to the original model to secure a margin for modeling errors and noises. This additional time delay is also applied as a time advance as in the zero phase approximation case.

3.2. Attainable bandwidth for convergence of ILC by two stable approximations

Using the zero phase approximation and the time delay approximation, non-minimum phase zeros are replaced by stable poles or stable zeros for the model inversion. Each inversion-stable model approximation provides benefits in a phase or magnitude error, but the accuracy of the both approximations is band-limited. Because of the large approximation errors at high frequencies, a Q filter is designed as a low-pass filter to suppress them for the ILC convergence. The bandwidth of the Q filter is also important for ILC performance because it also defines a correction bandwidth of the ILC. In this subsection the attainable bandwidth of the both inversion-stable model approximations w_c is analyzed to evaluate attainable ILC performance of the both approximations.

The maximum attainable bandwidth w_c is defined by the upper-bound of a set $w \in [0, w_c)$ in which the ILC convergence criterion of Eq. (6) is satisfied. For defining this bandwidth, an ideal low-pass Q filter is assumed as

$$Q(w) = \begin{cases} 1, & w \in [0, w_c), \\ 0, & w \in [w_c, \infty). \end{cases} \tag{12}$$

The ideal Q filter selectively updates the ILC input from the errors in the bandwidth $[0, w_c)$ and ignores the errors out of this bandwidth.

For simplicity and to the link to the galvanometer model in Eq. (1), the attainable bandwidth is analyzed for non-minimum phase zeros of a second order system as

$$H(s) = \frac{K_h(s^2 + 2\zeta_h\omega_h s + \omega_h^2)}{s^2} = \frac{B_H^s(s)B_H^u(s)}{A_H(s)}, \tag{13}$$

where K_h is an overall gain of the system, $\zeta_h \in R^-$ and ω_h denote the damping parameter and the natural frequency of non-minimum phase zeros, respectively. By the same way of Eq. (9), the system can be factorized by the polynomials of the non-minimum phase zeros, i.e. $B_H^u(s) = s^2 + 2\zeta_h\omega_h s + \omega_h^2$, the minimum phase zeros $B_H^s(s) = K_h$, and the denominator $A_H(s) = s^2$.

Laplace domain is used for the analysis without loss of generality. Then the zero phase approximation can be written as (Park, Hun Chang, & Lee, 2003):

$$\hat{H}_{zp}(s) = \frac{B_H^s(s)}{A_H(s)} \frac{(B_H^u(0))^2}{B_H^u(-s)}. \tag{14}$$

Due to zero phase error, i.e. $\left|\arg(H(w)) - \arg(\hat{H}_{zp}(w))\right| = 0$, the approximation error is given only in magnitude. From Eqs. (12), (7), and $L(w) = \rho\hat{H}_{zp}(w)$, the attainable bandwidth $w_{c,ZP}$ is defined by an upper bound of frequency that satisfies the convergence of ILC as

$$\rho\left|\frac{H(w_n)}{\hat{H}_{zp}(w_n)}\right| = \rho\frac{\left|B_H^u(w_n)\right|^2}{\left|B_H^u(0)\right|^2} < 2, \quad \forall\, w_n \in [0, \;\; w_{c,ZP}/\omega_h), \tag{15}$$

where w_n denotes a relative frequency with respect to the natural frequency of the non-minimum phase zeros, $w_n = w/\omega_h$. Since Eq. (15) is satisfied at $w_n = 0$, the learning gain has to be smaller than 2. Substitute Eq. (15) with Eq. (13), the attainable ILC bandwidth $w_{c,zp}$ of zero phase approximation is obtained as

$$w_{c,ZP}/\omega_h = \sqrt{1 - 2\zeta_n^2 + \sqrt{(1 - 2\zeta_n^2)^2 + 2\rho^{-1} - 1}}. \tag{16}$$

The bandwidth of the time delay approximation can be obtained in the similar manner. For simplicity, zero magnitude error tracking control (ZMETC) is used, i.e. $\hat{B}^u(s) = B^u(-s)$ (Butterworth et al., 2012). Then the time delay approximation can be written as

$$\hat{H}_{td}(s) = \frac{B_H^s(s)B_H^u(-s)e^{-t_h s}}{A_H(s)}, \tag{17}$$

where t_h denotes an additional delay for the phase correction. Since there is no magnitude error, i.e. $\left|H(w)/\hat{H}_{td}(w)\right| = 1$, only the phase error influences the convergence of ILC. From Eqs. (17) and (13), (8) can be rewritten as

$$\left|\arg(H(w_n)) - \arg(\hat{H}_{td}(w_n))\right| = \left|2\arctan\left(\frac{2\zeta_h w_n}{1 - w_n^2}\right) + t_h\omega_h w_n\right| < \arccos\left(\frac{\rho}{2}\right) \le \frac{\pi}{2}. \tag{18}$$

For the analysis, an appropriate time delay t_d^* has to be chosen to maximize the attainable bandwidth $w_{c,TD}$ subject to (18) for $w_n \in [0, w_{c,TD}/\omega_h)$. To confine the candidates of t_h^*, at least one of the critical points of the phase in the bandwidth are tangent to the boundary of the stability criterion Eq. (18). In other words, the appropriate time delay t_h^* can be obtained by

$$0 = \frac{d}{dw_n}\{\arg(H(w_n)) - \arg(\hat{H}_{td}(w_n))\}$$

$$\leftrightarrow t_h^* = -\frac{4(1 + \bar{w}_n^2)\zeta_h}{\omega_h(1 + \bar{w}_n^4 + \bar{w}_n^2(-2 + 4\zeta_h^2))}, \tag{19}$$

where $\bar{w}_n \in \mathbb{R}^+$ is the solution of the active constraints of Eq. (18), i.e.

$$\left|2\arctan\left(\frac{2\zeta_h\bar{w}_n}{1 - \bar{w}_n^2}\right) + t_h^*\omega_h\bar{w}_n\right| = \arccos\left(\frac{\rho}{2}\right). \tag{20}$$

Since the critical points by Eq. (19) can be multiple and there are two equality conditions from Eq. (20), multiple candidates of $\bar{w}_n$ and t_h^* may be obtained. From the candidates, the t_h^* is chosen that provides the maximum attainable bandwidth that satisfies Eq. (18) for $w_n \in [0, w_{c,TD}/\omega_h)$. Since there is no closed form of the solution of t_h^* available, the attainable bandwidth of the time delay approximation can be numerically obtained for each ζ_h.

As observed from Eqs. (16) and (18), the relative attainable bandwidth w_c/ω_h is a function of the damping parameter ζ_h and the learning gain ρ. Fig. 10 compares the relative attainable bandwidths of the zero phase approximation of Eq. (16) (dashed lines) and time delay approximation of Eq. (18) (solid lines) with respect to ζ_h and ρ. For low damped non-minimum phase zeros, the attainable bandwidth of time delay approximation is smaller than the zero phase approximation due to abrupt phase errors around the natural frequency ω_h. For $\rho = 1$ and the non-minimum phase zeros with $|\zeta_h| > 0.21$, the time delay approximation (black solid line) provides a wider bandwidth than the zero phase

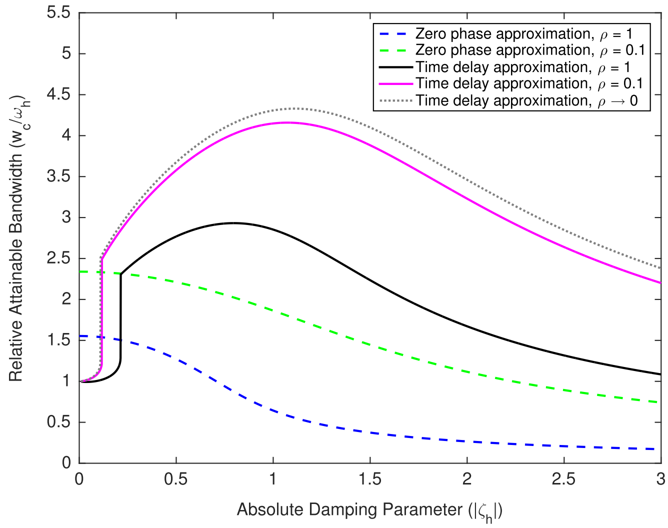


Fig. 10. Relative attainable bandwidths w_c/ω_h of the zero phase approximation (dashed lines) and time delay approximation (solid lines) with respect to the damping parameter ζ_h and the learning gain ρ of 1 and 0.1 (represented in different colors). The upper bound of the attainable bandwidth for time delay approximation ($\rho \approx 0$) is shown by a gray dotted line. For $\rho = 1$ and $|\zeta_h| > 0.21$, the time delay approximation can provide a wider bandwidth than the zero phase approximation.

approximation (blue dashed line).

By decreasing the learning gain ρ, the bandwidth of ILC by both approximations can be extended. Fig. 10 also shows larger attainable bandwidths of the zero phase approximation (green dashed line) and the time delay approximation (magenta solid line) with a small learning gain of ρ=0.1. The lower bound of $|\zeta_h|$ for a wider bandwidth of the time delay approximation than the zero phase approximation also reduces to 0.12. About $\rho \approx 0$, there is an upper bound of the attainable bandwidth (gray dotted line) for the time delay approximation, while zero phase approximation does not have such an upper bound. However, choosing a too small learning gain slows down the learning speed drastically. For $\rho > 0.1$ and for a large enough absolute damping $|\zeta_h| > 0.12$, the time delay approximation provides a wider ILC bandwidth than the zero phase approximation. From the damping parameter of a non-minimum phase zero pair in the system, a better stable approximation can be selected for a wider ILC control bandwidth based on Fig. 10.

This analysis can be easily extended to a single real non-minimum phase zero or higher order non-minimum phase zeros (Åström & Hägglund, 2006; Skogestad & Postlethwaite, 2005). For the high order case, however, the attainable bandwidth usually shrinks since the stability condition stays the same. In practical design, the actually attainable bandwidth is usually smaller than the limit of each approximation due to the design margin for the learning speed of the ILC and the modeling errors, uncertainties and noise in the real system.

3.3. Iterative learning control design

Since the stabilized galvanometer scanner model $P(z)$ has a pair of non-minimum phase zeros, the model is approximated by the stable inversion methods. Zero phase approximation ILC_{ZP} and the stable inversion with an input delay ILC_{TD} are investigated in this section. The ILC design is divided into two steps: stable learning filters L for the galvanometer scanner and Q filter design based on the modeling error.

For the design of inversion-based ILCs, the learning filters are designed by two different stable inversion methods. ILC_{ZP} is designed by Eq. (10) from the model with non-minimum phase zeros. For the conversion of continuous time model $\hat{P}(s)$ to the discrete time model $\hat{P}(z)$, Tustin's method is used since the stable region of the s-plane is exactly mapped into the stable region of the z plane. The learning filter of the zero phase approximation is obtained by Eq. (10), i.e. $L_{ZP}(z) = \hat{P}_{zp}^{-1}(z)$. ILC_{TD} uses the time delay approximation model with only minimum phase zeros in Eq. (11), i.e $L_{TD}(z) = \hat{P}_{td}^{-1}(z)$. This time delay approximation is designed in the continuous time domain $\hat{P}(s)$ and the approximated continuous time model is converted to the discrete time model by Tustin's method as ILC_{ZP}. The learning gain ρ is set to 1 in both cases.

For the Q filter design of each ILC, a zero-phase 8th order Butterworth IIR filter is used. The frequency response of the robustness filter $Q(z)$ is designed to satisfy the convergence criterion (6) for all frequencies. In the case of the galvanometer model of Eq. (1), with non-minimum phase zeros of the damping parameter of −0.85 at the frequency of 25 kHz, Fig. 10 shows that ILC_{TD} with ρ=1 has a bandwidth up to 73.2 kHz, larger than ILC_{ZP} with a bandwidth up to 20.2 kHz. The bandwidth of the implemented ILC is lower than this limit due to the modeling error, which become significant for the higher frequencies than 42 kHz as Fig. 5 shows. Based on the analysis and the measured Bode plot, the cutoff frequencies of the Q-filters for ILC_{ZP} and ILC_{TD} are determined as 20 kHz and 42 kHz, respectively, as marked in Fig. 9.

This shows that ILC can have a wider control bandwidth than the frequency of the non-minimum phase zeros while the bandwidth of a feedback controller is usually limited by the frequency of the non-minimum phase zeros (Skogestad & Postlethwaite, 2005). In the following section, benefits of the wide control bandwidths of ILC are demonstrated by the tracking performances in comparison with a commercial feedback controller.

4. Experimental results

4.1. Implementation of ILC

For ILC implementation for the galvanometer scanner, a data acquisition system consisting of a DAQ card (DAQ-2010, ADLink Technology Inc., New Taipei City, Taiwan) is used. The measurement data are processed in Labview (National Instruments Co., Austin, Texas, USA) that is capable of the seamless streaming of the scanning signal (Fantner et al., 2005) while calculating the updated scanning signal via ILC in Matlab. The angle of the mirror is sampled and recorded from the encoder output at a sampling frequency f_s of 526 kHz, i.e. the sampling time T_s of 1.9 µs. Accordingly, the delays in the discrete time learning filters are set as k_d=1, and k_a=9 samples.

For output measurements for the ILC update, the encoder signals of 512 triangular scans are averaged to reduce the influence of measurement noise (Schitter, 2009). The reference trajectory y^{ref} is used for the initial ILC input u^0, and the learning of ILC stops at the 20th iteration, when ILC is already converged to its error limit.

4.2. Preparation of the triangular scanning reference

A triangular scanning reference is desirable for a fast scanning, however, it include high frequency components due to the sharp turnaround (Fleming & Wills, 2009; Schmidt et al., 2014), which can lead to errors in the active scan area and a large input peak (Fleming & Wills, 2009). To reduce the unwanted high frequency errors, input-shaping methods (Singer & Seering, 1988; Singhose, 2009) can be used. For high speed atomic force microscopy, an input shaping method reduces the resonance in the active scan area and results in less image distortion (Schitter, Thurner, &

Hansma, 2008). For the galvanometer scanner, polynomials and sinusoidal functions are studied for the effective turnaround (Duma, 2010).

To reduce the high frequency content of the reference signal, an input shaping filter is implemented in the form of a finite impulse response (FIR) filter for the triangular scanning reference (Fleming & Wills, 2009; Singer & Seering, 1988). FIR filters have desirable properties such as a linear phase all over the frequency and a constant group delay, which can be easily corrected by zero phase filtering in advance. The innate finite response structure, only finite number of samples near the turnaround is smoothened while the active scanning region is kept linear.

In the experiments, a 5th order FIR filter is used as follows:

$$H(z) = \tfrac{10}{32}z^{-3} + \tfrac{5}{32}z^{-2} + \tfrac{1}{32}z^{-1} + \tfrac{1}{32}z^{0} + \tfrac{5}{32}z^{1} + \tfrac{10}{32}z^{2}. \tag{21}$$

This FIR filter has the first zero at 63 kHz and the magnitude is -6.3 dB at 42 kHz. This FIR filter only change ± 5 samples of the linear scan region near the turnaround.

For evaluation of ILC three pre-filtered triangular reference signals with a scanning rate of 1.03 kHz (512 samples per a triangular scan), 2.06 kHz (256 samples per a triangular scan) and 4.11 kHz (128 samples per a triangular scan) are examined, which corresponds to 2056, 4112, and 8224 lines per second for scanning laser microscopes, respectively. The scanning amplitudes in optical angle are 0.235°, 0.185° and 0.111°, respectively.

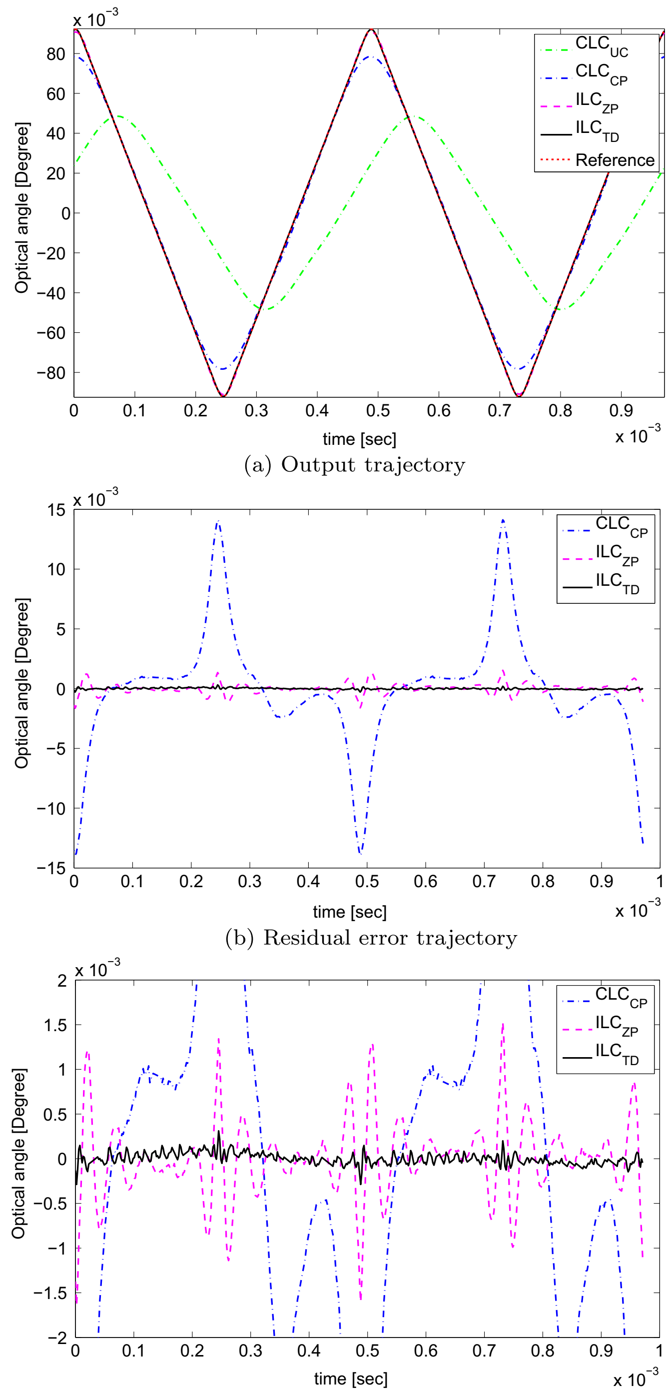
4.3. Tracking results

Experimental results of the designed ILC_{ZP} and ILC_{TD} are presented for the galvanometer scanner and compared with the closed-loop controller (CLC_{UC}, CLC_{CP}) of the conventional servo driver.

Fig. 11 shows the scanning angle and error trajectory for each controller at the 2.06 kHz scan rate, i.e. 4112 lines per second. Fig. 11(a) shows the final angle trajectory of ILC_{TD} (black solid line), ILC_{ZP} (magenta dashed line), and CLC_{UC} (green dash-dot line). First, both ILCs automatically track the trajectory without any phase lag and magnitude error while the CLC_{UC} has about a 74 μs time lag and 40% gain mismatch. Both ILC can compensate this phase and gain mismatch while the performance of ILC can outperform the conventional feedback controller, even when this gain and phase margin is compensated by prior knowledge in advance (Trepte & Liljeborg, 1994). Therefore, the closed loop controller with the phase and magnitude mismatch compensation, denoted by CLC_{CP} (blue dash-dot line in Fig. 11(a)), is used for further comparison. The gain and phase lag are compensated based on the position and average gradient of 11 data points that are centered closely to the zero degree line, denoting the center of the scanning laser microscope image. Assuming that the system is linear, CLC_{CP} obtains the same results shifting the reference and gain adjustments.

Though the phase and gain errors are compensated for CLC_{CP}, ILCs outperform CLC_{CP} with a better tracking performance by a smaller RMS error as well as a smaller peak-to-peak tracking error. Fig. 11(b) shows that the error trajectory of ILC_{TD} and ILC_{ZP} are much smaller than CLC_{CP}. The high bandwidth of ILC_{TD}, as discussed in Section 3, leads to the smallest error, followed by ILC_{ZP} and the largest error for CLC_{CP} in all cases. The peak-to-peak errors of ILC_{TD} and ILC_{ZP} are up to 46 and 8 times smaller than that of CLC_{CP}, respectively. For the RMS errors the improvements are even better. At 2.06 kHz scan rate, the RMS error of ILC_{TD} is 73 times smaller than that of CLC_{CP}. ILC_{ZP} shows less improvement of tracking performance than ILC_{TD}, but its RMS errors are also up to 10 times smaller than that of CLC_{CP}.

Not only does ILC result in smaller errors, but it also enables the rejection of periodic disturbance and nonlinearity compensation.



(a) Output trajectory

(b) Residual error trajectory

(c) Zoomed residual error trajectory

Fig. 11. Output and error trajectories of ILC_{TD} (black solid line), ILC_{ZP} (magenta dashed line), CLC_{CP} (blue dash-dot line) and CLC_{UC} (green dash-dot line) at a 2.06 kHz scanning frequency (4112 lines per second). The error trajectory is zoomed to better compare the error between ILC_{TD} and ILC_{ZP}.

The control signals as shown in Fig. 12 illustrate that the final inputs between the trace and the retrace scan region are asymmetric. This indicates that there are nonlinearities or a periodic disturbance in the galvanometer actuation. Besides, the error trajectories of the CLC_{CP} in Fig. 11(b) illustrate the asymmetrical residual errors between the trace and the retrace scan region, which leads to a misalignment of the individual lines in the image. Both ILC, ILC_{TD} and ILC_{CP}, compensate for such errors between trace and retrace, allowing doubling the speed by the bidirectional scan,

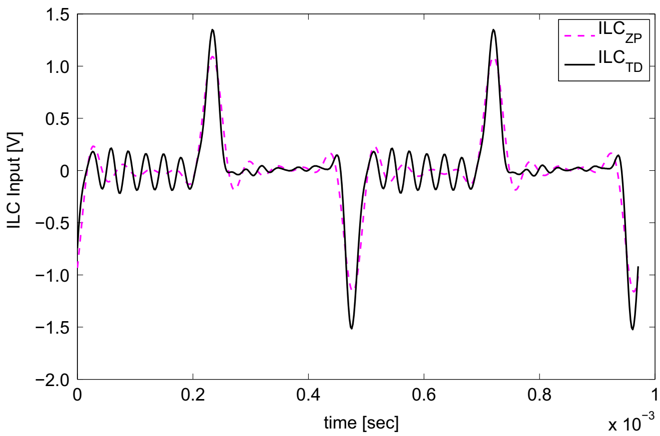


Fig. 12. Input trajectories of ILC_{TD} (black solid line) and ILC_{ZP} (magenta dashed line) at a 2.06 kHz scanning frequency.

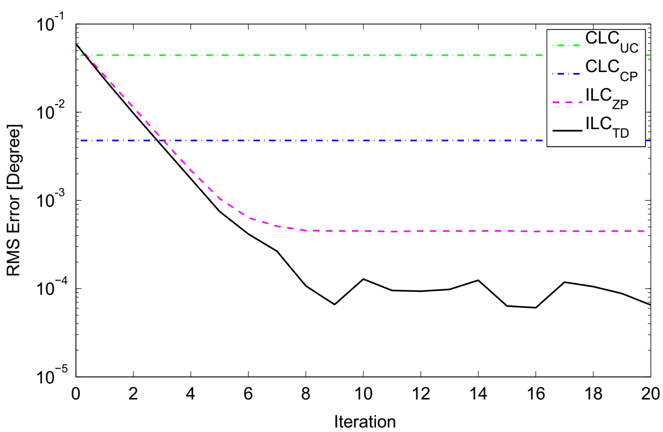


Fig. 13. RMS error as a function of ILC iterations.

Table 2
The final RMS error (e_{rms}) and peak-to-peak error (e_{pp}) in percentage (%) for different controllers, taken with respect to the amplitude of the reference triangle command at three different scan rates.

e_{rms} (e_{pp})	1.03 kHz	2.06 kHz	4.11 kHz
CLC_{UC}	15.84 (56.48)	24.04 (58.41)	38.22 (53.84)
CLC_{CP}	0.64 (4.85)	2.59 (11.93)	5.29 (11.70)
ILC_{ZP}	0.090 (0.88)	0.24 (1.35)	0.83 (2.02)
ILC_{TD}	0.026 (0.13)	0.035 (0.25)	0.11 (0.36)

since the other dynamics in beam scanning are negligible besides the galvanometer scanner, and the PSF is symmetric during the scanning.

Fig. 13 illustrates the evolution of the RMS error along the iteration axis for ILC at 2.06 kHz scan rate. Four to five more iterations are required for ILC_{TD} to reach the final value in all cases than for ILC_{ZP}, but the convergence speed of both ILC is similar and ILC_{TD} can attain the smaller RMS error. The RMS errors of the CLC_{UC} and CLC_{CP} are given for comparison. The final RMS errors and peak-to-peak errors of 1.03 kHz, 2.03 kHz, and 4.03 kHz are summarized in Table 2. For the case of ILC_{TD} scanning at 4112 lines per second the RMS error is a factor of 73 smaller than controlling the galvanometer scanner with the conventional feedback controller, which clearly demonstrates that ILC enables scanning at a high speed with a significantly reduced tracking error.

5. Conclusion

Iterative learning control (ILC) schemes are proposed for a galvanometer scanner in scanning laser microscopy, achieving a fast, linear, and accurate bidirectional scanning. Two inversion types of ILC, a zero phase approximation ILC_{ZP} and a delay approximation ILC_{TD}, are designed based on the linear model of the stabilized galvanometer scanner system with non-minimum phase zeros. The attainable bandwidths of both approximations are analyzed for a complex pair of non-minimum phase zeros, which shows that the time delay approximation can provide a wider bandwidth of ILC than the zero phase approximation if the absolute value of the damping parameter of the non-minimum phase zeros is large enough. In the implementation, the bandwidths of ILC_{ZP} and ILC_{TD} can be set up to 20 kHz and 42 kHz, respectively, considering the convergence of ILC. The designed ILCs are evaluated using a high performance galvanometer scanner at high scanning speed, and are compared to a conventional closed loop controller. Both ILCs outperform the conventional closed loop control in terms of the root mean square (RMS) error and peak-to-peak tracking error. ILC_{TD} show an RMS error that is up to 73 times smaller as compared to the gain and phase corrected closed loop control CLC_{CP} of the commercial system. ILC_{ZP}, which has a narrower control bandwidth than ILC_{TD}, outperforms the CLC_{CP} by an up to 10 times smaller RMS error. This result clearly demonstrates the potential of ILC for fast and accurate motion control of the galvanometer scanner at scanning speeds of up to 8000 lines per second, enabling distortion free imaging at improved temporal resolution in conventional scanning laser microscopy.

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